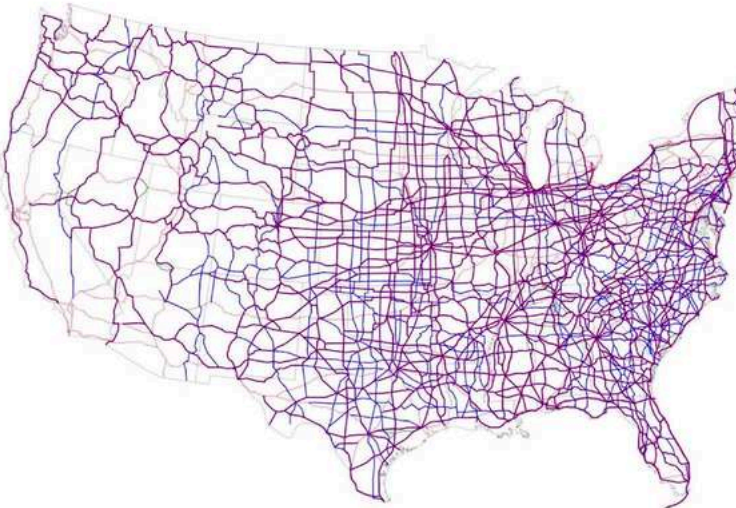


L1: Examples of Systems Studied by Network Scientists



2 videos

Examples of Systems ...

6:17

Examples of Systems ...

5:59

Highway Network

Georgia Tech

0:00 / 6:17

CC

1x

In the next few pages, I would like to give you some examples of the problems we will study during the semester in this course. The description will be high-level as I only want to give you a taste of what is coming up next.

Source research articles for the images in these videos

- [Chains of Affection: The Structure of Adolescent Romantic and Sexual Networks](https://www.cis.upenn.edu/~mkearns/teaching/NetworkedLife/teensex.pdf) 
(<https://www.cis.upenn.edu/~mkearns/teaching/NetworkedLife/teensex.pdf>)

- [Rise of China in the International Trade Network: A Community Core Detection Approach](https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4138169) ➞ (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4138169)
- [Predicting the Fission Yeast Protein Interaction Network](https://pdfs.semanticscholar.org/0595/6042c6bd23eb49b5071964ce2d04edb26921.pdf?_ga=2.238092041.829074902.1579584841-796840842.1579584841) ➞ (https://pdfs.semanticscholar.org/0595/6042c6bd23eb49b5071964ce2d04edb26921.pdf?_ga=2.238092041.829074902.1579584841-796840842.1579584841)
- [Networking Our Way to Better Ecosystem Service Provision](https://www.sciencedirect.com/science/article/pii/S0169534715003006) ➞ (https://www.sciencedirect.com/science/article/pii/S0169534715003006)
- [Influence of fake news in Twitter during the 2016 US presidential election](https://www.nature.com/articles/s41467-018-07761-2) ➞ (https://www.nature.com/articles/s41467-018-07761-2)

Also, some images are copied from the following online sources:

- [Wireless Data Center with Millimeter Wave Network](https://ieeexplore.ieee.org/document/5684121) ➞ (https://ieeexplore.ieee.org/document/5684121)
- [Data visualization for social network analysis](https://cambridge-intelligence.com/use-cases/social-networks/) ➞ (https://cambridge-intelligence.com/use-cases/social-networks/)
- [The bottlenose dolphin community of Doubtful Sound features a large proportion of long-lasting associations](https://link.springer.com/article/10.1007/s00265-003-0651-y) ➞ (https://link.springer.com/article/10.1007/s00265-003-0651-y)
- [The synapse, Khan Academy](https://www.khanacademy.org/science/biology/human-biology/neuron-nervous-system/a/the-synapse) ➞ (https://www.khanacademy.org/science/biology/human-biology/neuron-nervous-system/a/the-synapse)
- [Action potentials and synapses, The University of Queensland](https://qbi.uq.edu.au/brain-basics/brain/brain-physiology/action-potentials-and-synapses) ➞ (https://qbi.uq.edu.au/brain-basics/brain/brain-physiology/action-potentials-and-synapses)
- [Alterations in Brain Network Topology and Structural-Functional Connectome Coupling Relate to Cognitive Impairment](https://www.frontiersin.org/articles/10.3389/fnagi.2018.00404/full) ➞ (https://www.frontiersin.org/articles/10.3389/fnagi.2018.00404/full)
- [The Measurement Standard, Carma](https://persiababylonia.org/archives/methods/social-network-analysis-and-cuneiform-archives/) ➞ (https://persiababylonia.org/archives/methods/social-network-analysis-and-cuneiform-archives/)
- [Schizophrenia interactome with 504 novel protein–protein interactions](https://www.nature.com/articles/npjSchz201612%E2%80%8B) ➞ (https://www.nature.com/articles/npjSchz201612%E2%80%8B)